

DAY 4

1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

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GPU Brand: AMD

GPU Model: AMD Radeon Graphics (Integrated with Ryzen 5 7000 Series)

Type: Integrated GPU

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



CPU Cooler (Heatsink & Fan): Draws heat directly away from the processor using a metal baseplate and conductive heat pipes, then disperses it into the air.

Case Fans: Pull cool air into the chassis (intake) and exhaust hot air out to maintain a stable internal temperature.

GPU Fans: Built directly onto the graphics card to cool its dedicated processor and memory under heavy graphical loads.

Liquid Cooling Radiator (Alternative): Uses circulating liquid to pull heat away from the CPU and moves it to a radiator, where fans push the heat out of the PC case.

3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Feature	Air Cooling	Liquid Cooling
Cost	Low cost, affordable and commonly used in most GPUs	High cost due to pumps, radiators, and complex design

Feature	Air Cooling	Liquid Cooling
Cooling Efficiency	Moderate efficiency; can struggle with very high-performance GPUs	Very high efficiency; better heat removal for powerful GPUs
Maintenance	Low maintenance; just occasional dust cleaning required	Higher maintenance; risk of leaks and pump failure checks needed

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

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For a PC handling graphics-heavy gaming and high-resolution streaming, choosing liquid cooling (AIO) for the GPU is ideal.

Intense workloads generate massive amounts of thermal energy, and liquid coolant is denser and more efficient at absorbing heat than pure air. It moves the heat directly to a radiator, keeping peak temperatures lower over long, continuous streaming and gaming sessions. This prevents thermal throttling and keeps fan noise to an absolute minimum while maintaining peak frame rates.